TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Electronics: Rectifiers With and Without Filter, and the Transistor Amplifier

A diode passes one half of the a.c. cycle and blocks the other — that single fact builds every d.c. power supply. A transistor does the opposite trick: a small signal at its base commands a large copy at its collector.

EEE-2204 · ELECTRICAL & ELECTRONICS SESSIONAL · EXPERIMENT 4 · LEVEL 2 TERM 2

1. Aim and theory

Aim: (i) to study half-wave and full-wave (bridge) rectifiers, recording the output waveforms with and without a capacitor filter on the oscilloscope, and to measure the d.c. output in each case; (ii) to study a single-stage common-emitter transistor amplifier and measure its voltage gain.

Rectifiers. A p–n diode conducts only when forward biased. In the **half-wave** rectifier one diode passes the positive half-cycles and blocks the negative ones, so the load receives pulses at the supply frequency and the average (d.c.) output is

$$V_{dc} = V_m / \pi = 0.318 \cdot V_m \quad \text{(half-wave)}$$

The **full-wave bridge** uses four diodes in a diamond: on each half-cycle one diagonal pair conducts and steers the current through the load in the *same* direction, so both halves of the wave are used, the ripple frequency doubles to 100 Hz, and

$$V_{dc} = 2V_m / \pi = 0.637 \cdot V_m \quad \text{(full-wave)}$$

The unfiltered output is pulsating d.c. — its quality is measured by the **ripple factor** (r.m.s. ripple ÷ d.c. value): 1.21 for half-wave, 0.482 for full-wave. A large **capacitor filter** across the load charges up to near the peak V_m at each pulse and discharges slowly through the load between pulses, filling the gaps: the output becomes nearly steady d.c. with a small sawtooth ripple of peak-to-peak value approximately $V_{r(pp)} \approx I_{dc}/(f \cdot C)$ for half-wave and $I_{dc}/(2f \cdot C)$ for full-wave — the bridge with a filter is the standard power-supply front end.

Transistor amplifier. In the common-emitter stage, a voltage divider (R_1, R_2) holds the base about 0.7 V above the emitter resistor so a steady bias current flows. A small input signal, coupled in through a

capacitor, wiggles the base voltage; the collector current copies the wiggle multiplied by β, and the resulting swing across R_C is the amplified — and inverted — output. With the emitter resistor bypassed by a capacitor, the mid-frequency voltage gain is approximately

$$A_v = -R_C / r_e' \quad \cdot \quad r_e' = 25\ \text{mV} / I_E \quad \cdot \quad \text{gain in dB} = 20 \cdot \log_{10} |A_v|$$

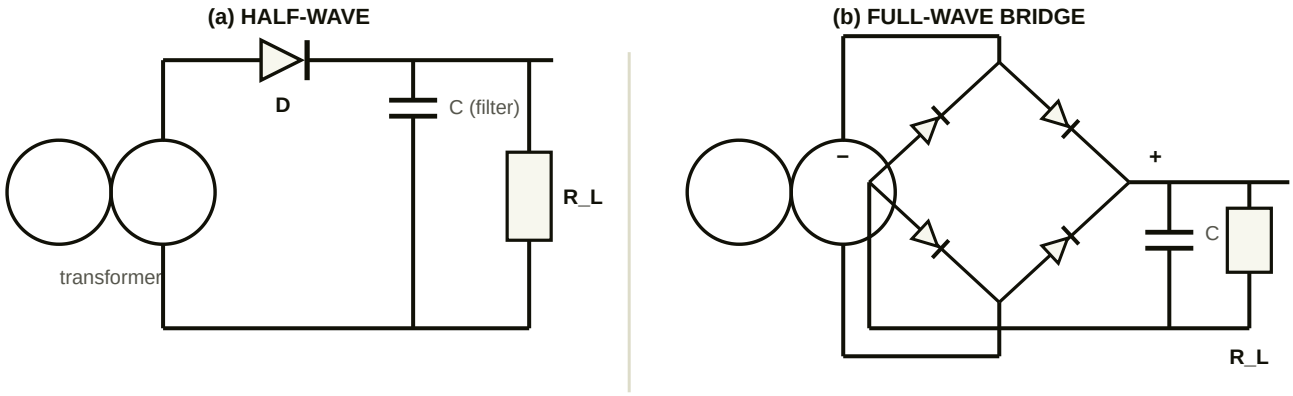
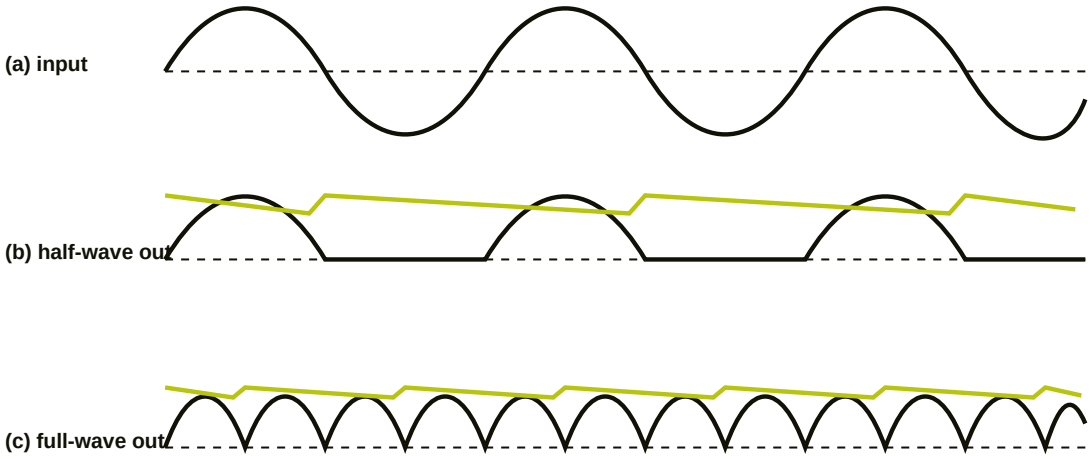


Figure 1 — The two rectifier circuits. (a) Half-wave: one diode in series with the load; the filter capacitor C is connected across R_L for the second part of the test. (b) Bridge: the a.c. enters the top and bottom corners of the diode diamond and the load connects between the right (+) and left (−) corners — whichever half-cycle is live, the load current keeps the same direction.



black = without filter · green = with capacitor filter (charges to the peak,

Figure 2 — Oscilloscope waveforms. (a) The sinusoidal input. (b) Half-wave output: alternate half-cycles missing; the filter lifts the output to near the peak and it sags once per cycle. (c) Full-wave output: every half-cycle present, ripple at twice the frequency and — with the filter — half the sag.

2. Apparatus and procedure

- **Apparatus:** step-down transformer (230/12 V), diodes (1N4007) ×4, load resistor, electrolytic capacitor 1,000 µF, oscilloscope, a.c. and d.c. voltmeters; for the amplifier: transistor (BC547 or

similar), base and collector resistors, coupling and bypass capacitors, function generator, dual d.c. supply or battery pack.

- **Half-wave:** wire Figure 1(a) *without* the capacitor. Sketch the input and output waveforms from the oscilloscope; measure V_{dc} with the d.c. voltmeter and compare with $0.318 \cdot V_m$. Connect the capacitor across R_L and repeat — note how the trace changes and how V_{dc} climbs toward V_m.
- **Full-wave bridge:** wire Figure 1(b), checking each diode's polarity against the diagram before switching on. Repeat the same four observations; the ripple frequency on the trace should now be 100 Hz.
- **Amplifier:** build the common-emitter stage of Figure 3 with I_E set near 1 mA (check $V_E \approx 1$ V across $R_E = 1$ kΩ). Feed in a 1 kHz sine of 20 mV peak-to-peak, view input and output together, and measure the gain and the phase shift. Raise the input slowly and note the level at which the output begins to clip.

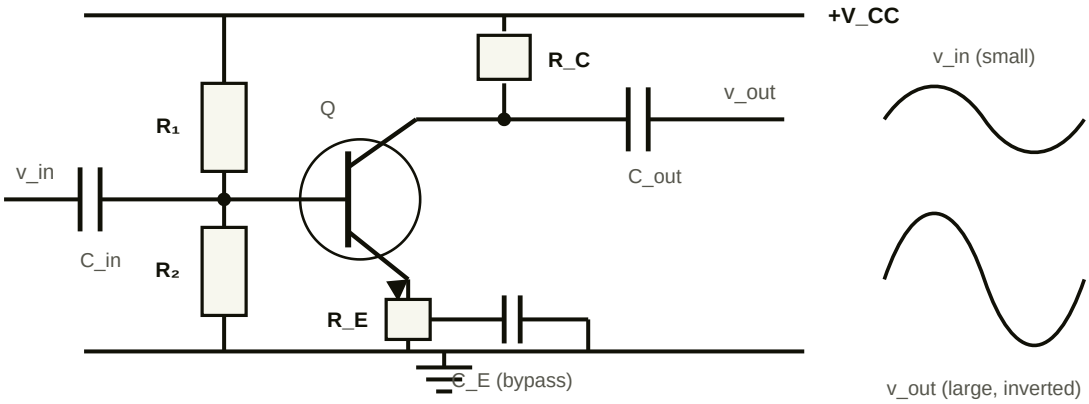


Figure 3 — Common-emitter amplifier with voltage-divider bias. R_1–R_2 set the base voltage, R_E stabilises the bias (its bypass capacitor C_E removes the a.c. feedback), and the signal collected at the collector through C_{out} is amplified and inverted, as the traces at the right show.

3. Worked calculation

Rectifiers. The transformer secondary is 12 V r.m.s., so $V_m = 12 \times \sqrt{2} = 17.0$ V.

$$\text{Half-wave: } V_{dc} = 0.318 \times 17.0 = 5.4 \text{ V} \quad \cdot \quad \text{Full-wave: } V_{dc} = 0.637 \times 17.0 = 10.8 \text{ V}$$

With a 1,000 µF filter fitted and the load drawing $I_{dc} = 100$ mA, the peak-to-peak ripple is

$$\text{Half-wave: } V_{r(pp)} = 0.100 / (50 \times 0.001) = 2.0 \text{ V} \quad \cdot \quad \text{Full-wave: } 0.100 / (2 \times 50 \times 0.001) = 1.0 \text{ V}$$

— the bridge halves the ripple for the same capacitor, one of the two reasons (with the doubled V_{dc}) it is the standard choice. The measured d.c. with filter sits close to V_m minus half the ripple, about 16 V here at light load.

Amplifier. The bias is set at I_E = 1 mA, so $r_e' = 25\ \text{mV} / 1\ \text{mA} = 25\ \Omega$. With R_C = 2 kΩ,

$$A_v = -R_C / r_e' = -2000 / 25 = -80 \quad \cdot \quad 20 \cdot \log_{10} 80 = 38\ \text{dB}$$

which matches the bench check: a 20 mV peak-to-peak input produced a 1.6 V peak-to-peak output (1.6/0.020 = 80), inverted with respect to the input. The minus sign is the 180° phase shift of the common-emitter stage.

4. Precautions

- Check every diode's polarity *before* switching on: a reversed diode in the bridge short-circuits the transformer on alternate half-cycles.
- Observe the electrolytic filter capacitor's polarity and voltage rating — reversed or over-voltaged electrolytics can vent or burst. Discharge it before rewiring.
- Remember the diode drop: measured V_{dc} falls about 0.7 V (half-wave) or 1.4 V (bridge, two diodes in the path) below the ideal formulas; silicon diodes are not ideal switches.
- On the oscilloscope, use a consistent volts/division and trigger level when sketching waveforms, and note the timebase — the 100 Hz ripple of the full-wave circuit is a key observation, not a detail.
- For the amplifier, verify the d.c. bias point (V_E, V_C) before applying any signal; a stage biased into saturation or cut-off clips no matter how small the input, and the "gain" measured from a clipped trace is meaningless.

Practice (viva preparation)

1. Why does the full-wave rectifier give twice the d.c. output of the half-wave rectifier for the same transformer?
2. A bridge rectifier is fed from a 9 V r.m.s. secondary. Estimate V_{dc} (a) without filter, (b) qualitatively with a large filter at light load.
3. What does the filter capacitor do between pulses, and why does doubling the capacitance halve the ripple?
4. The measured ripple in a full-wave supply is at 50 Hz, not 100 Hz. What fault do you suspect?
5. State two jobs of C_E and C_{out} in Figure 3, and why the output is inverted.

Answer key

1. The half-wave circuit uses only one half-cycle and discards the other; the bridge steers both half-cycles through the load in the same direction, so the average over a full cycle doubles: $2V_m/\pi$ against V_m/π.
2. $V_m = 9\sqrt{2} = 12.7$ V. (a) $V_{dc} = 0.637 \times 12.7 \approx 8.1$ V (about 6.7 V after two diode drops). (b) With a filter the output climbs toward the peak, roughly V_m minus the diode drops and half the ripple — around 11 V at light load.
3. It charges to near the peak at each pulse and then supplies the load current itself while the rectifier voltage falls below the capacitor voltage, discharging slowly. Ripple $\approx I \cdot \Delta t/C$, so doubling C halves the voltage sag for the same discharge time and current.
4. One diode of the bridge is open-circuit (or wrongly connected): the circuit has degenerated into a half-wave rectifier, giving one pulse per cycle — 50 Hz ripple.
5. C_E bypasses R_E for a.c. signals so the gain is not reduced by emitter degeneration, while R_E still stabilises the d.c. bias. C_{out} blocks the collector's d.c. level and passes only the a.c. output. Inversion: a rise in base voltage raises collector current, which drops more voltage across R_C, so the collector voltage falls — output moves opposite to input.

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